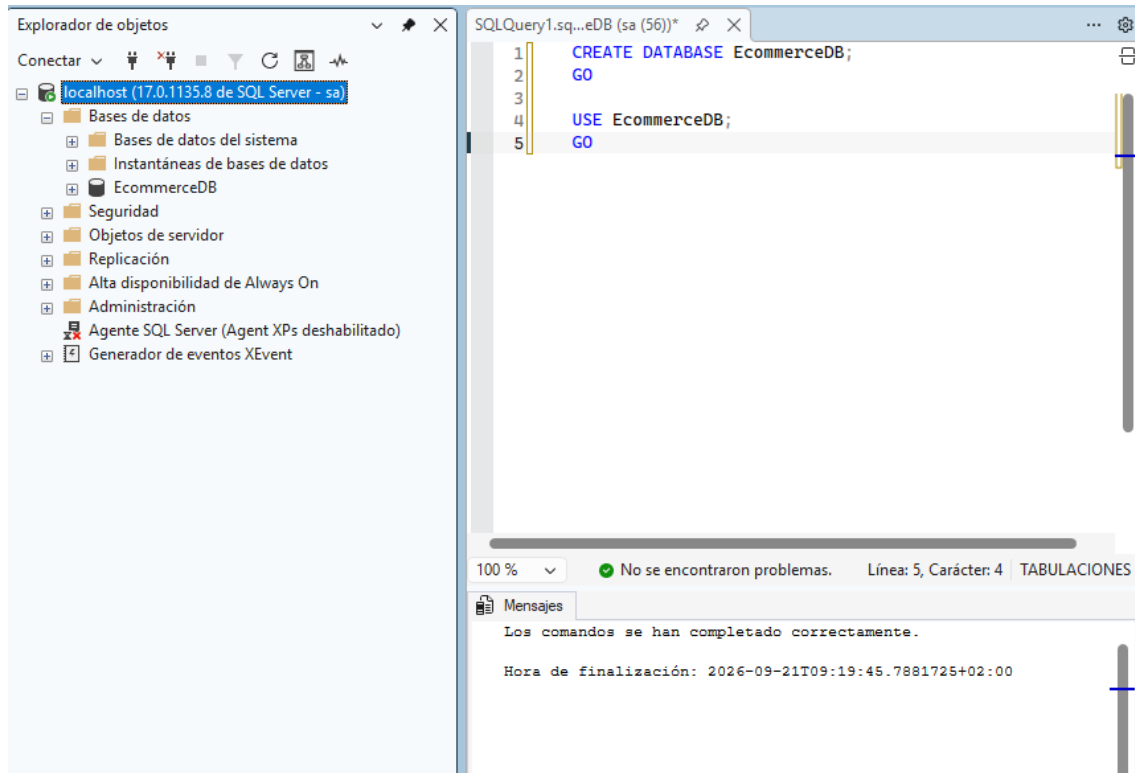


Contenido

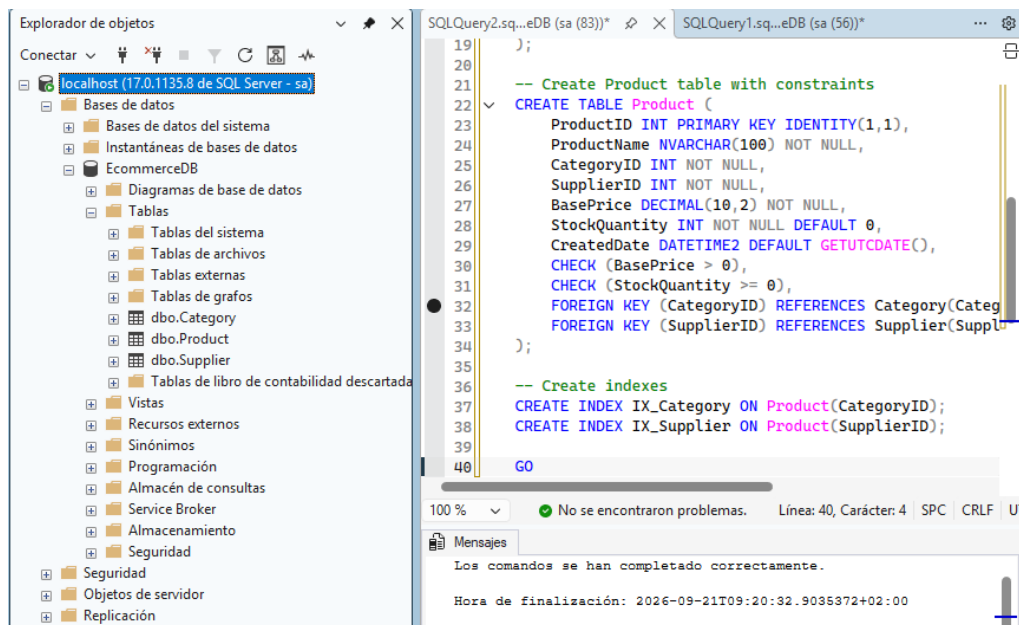
Creación de la bbdd	2
Creación de las tablas	3
Creación tablas	3
Insertar datos	3
Crea una tabla temporal para el historial de precios.	4
Consulta del historial	4
Agregar columnas JSON para metadatos.	5
Consulta de los datos json	5
Crear una tabla de pedidos particionada	6
Consulta de la partición	6
Crear detalles del pedido con SEQUENCE	7
Verificar los datos	7
Verificar objetos de la base de datos.....	8
Verificar el json y las consultas	8

Creación de la bbdd

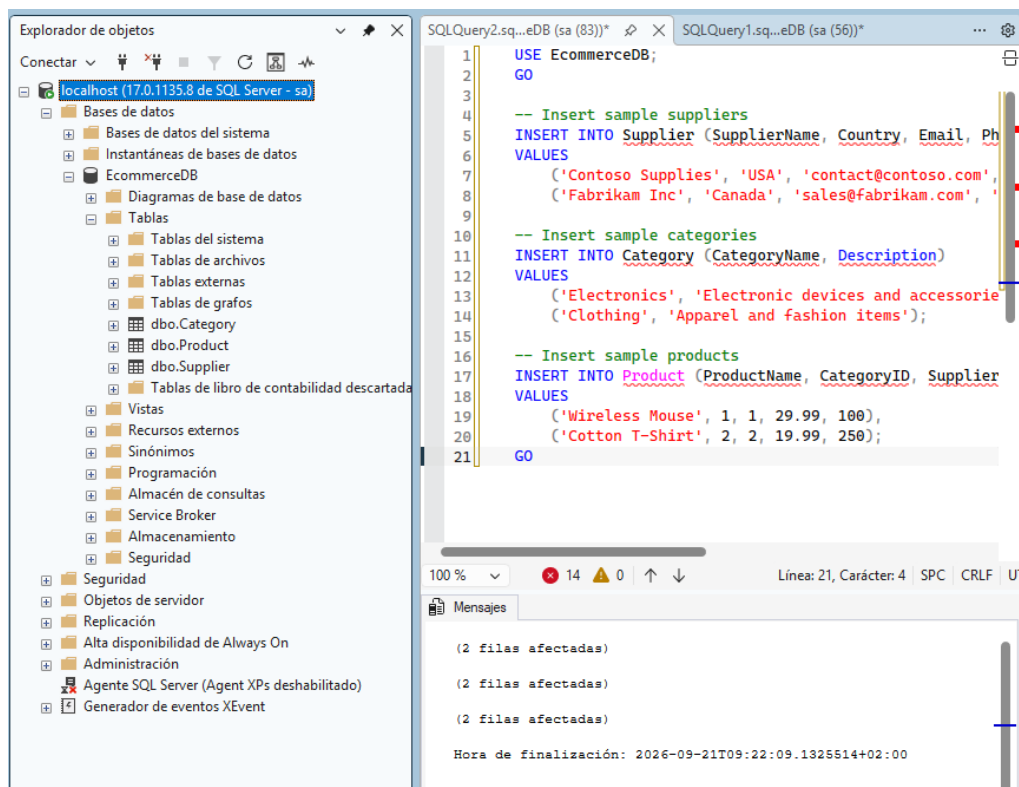


Creación de las tablas

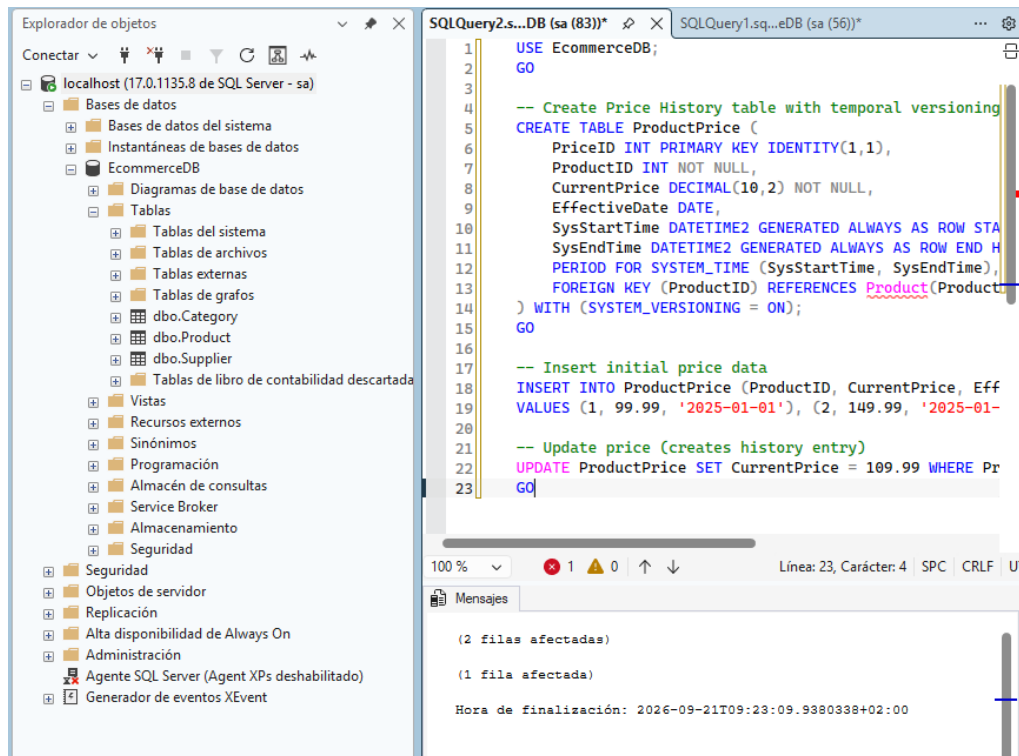
Creación tablas



Insertar datos



Crea una tabla temporal para el historial de precios.



The screenshot shows the SQL Server Enterprise Manager on the left, connected to localhost (17.0.1135.8 de SQL Server - sa). The right pane displays a SQL Query window for the 'EcommerceDB' database. The query creates a temporal table 'ProductPrice' with columns: PriceID (INT PRIMARY KEY IDENTITY(1,1)), ProductID (INT NOT NULL), CurrentPrice (DECIMAL(10,2) NOT NULL), EffectiveDate (DATE), SysStartTime (DATETIME2 GENERATED ALWAYS AS ROW START), and SysEndTime (DATETIME2 GENERATED ALWAYS AS ROW END). The table is configured with SYSTEM_VERSIONING = ON. The query also inserts initial price data and updates the current price for ProductID 1 to 109.99.

```
1 USE EcommerceDB;
2 GO
3
4 -- Create Price History table with temporal versioning
5 CREATE TABLE ProductPrice (
6     PriceID INT PRIMARY KEY IDENTITY(1,1),
7     ProductID INT NOT NULL,
8     CurrentPrice DECIMAL(10,2) NOT NULL,
9     EffectiveDate DATE,
10    SysStartTime DATETIME2 GENERATED ALWAYS AS ROW STA
11    SysEndTime DATETIME2 GENERATED ALWAYS AS ROW END H
12    PERIOD FOR SYSTEM_TIME (SysStartTime, SysEndTime),
13    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
14 ) WITH (SYSTEM_VERSIONING = ON);
15 GO
16
17 -- Insert initial price data
18 INSERT INTO ProductPrice (ProductID, CurrentPrice, Eff
19 VALUES (1, 99.99, '2025-01-01'), (2, 149.99, '2025-01-
20
21 -- Update price (creates history entry)
22 UPDATE ProductPrice SET CurrentPrice = 109.99 WHERE Pr
23 GO
```

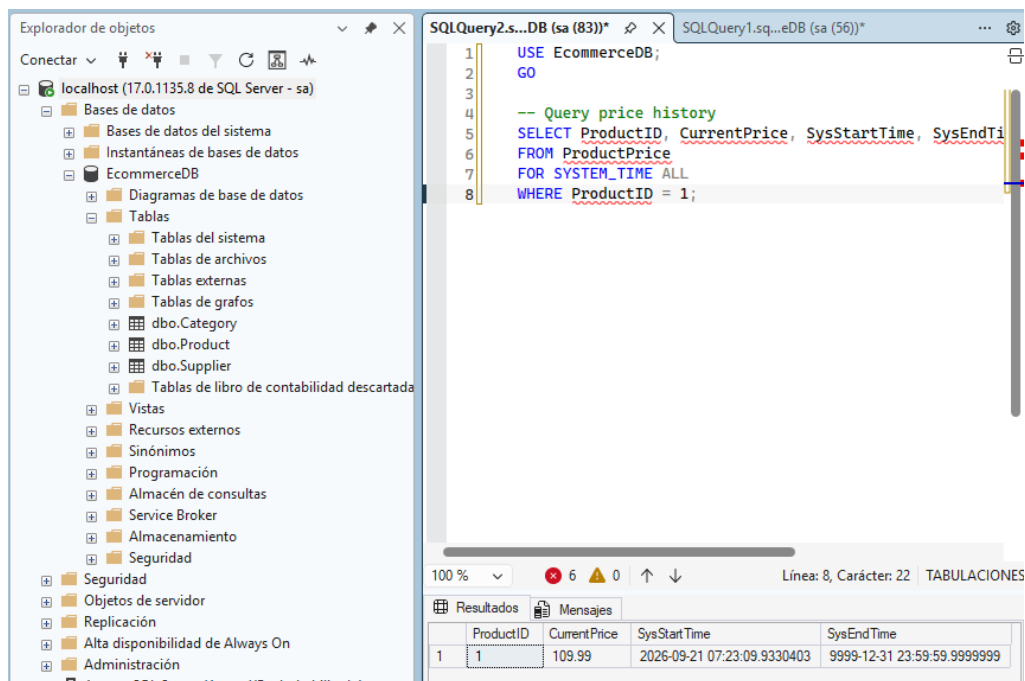
Mensajes

(2 filas afectadas)

(1 fila afectada)

Hora de finalización: 2026-09-21T09:23:09.9380338+02:00

Consulta del historial



The screenshot shows the SQL Server Enterprise Manager on the left, connected to localhost (17.0.1135.8 de SQL Server - sa). The right pane displays a SQL Query window for the 'EcommerceDB' database. The query selects the ProductID, CurrentPrice, SysStartTime, and SysEndTime from the ProductPrice table for ProductID 1.

```
1 USE EcommerceDB;
2 GO
3
4 -- Query price history
5 SELECT ProductID, CurrentPrice, SysStartTime, SysEndTi
6 FROM ProductPrice
7 FOR SYSTEM_TIME ALL
8 WHERE ProductID = 1;
```

Resultados

ProductID	CurrentPrice	SysStartTime	SysEndTime
1	109.99	2026-09-21 07:23:09.9330403	9999-12-31 23:59:59.9999999

Agregar columnas JSON para metadatos.

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Explorador de objetos' (Object Explorer) displays the database structure for 'localhost (17.0.1135.8 de SQL Server - sa)', including 'Bases de datos', 'EcommerceDB', and various system and user tables. The main window shows a SQL script being executed in 'SQLQuery2.s...DB (sa (83))'. The script includes the following commands:

```
1 USE EcommerceDB;
2 GO
3
4 -- Add metadata column to Product (JSON type requires
5 ALTER TABLE Product ADD Metadata JSON;
6 GO
7
8 -- Add computed column for indexing
9 ALTER TABLE Product ADD MetadataColor AS JSON_VALUE(Metadata, '$.color');
10 GO
11
12 -- Create index on the computed column
13 CREATE NONCLUSTERED INDEX IX_Product_Metadata_Color
14 ON Product (MetadataColor);
15 GO
16
17 -- Update products with metadata
18 UPDATE Product SET Metadata = N'{"color": "blue", "size": "large", "material": "cotton"}'
19 WHERE ProductID = 1;
20
21 UPDATE Product SET Metadata = N'{"color": "red", "size": "small", "material": "leather"}'
22 WHERE ProductID = 2;
23 GO
```

The status bar indicates 'Línea: 18, Carácter: 25'. The 'Mensajes' (Messages) pane at the bottom shows two warnings: 'Advertencia: la máxima longitud de clave para un índice no agrupado es de 32 bytes. El índice no agrupado puede no funcionar correctamente si se supera este límite.' (1 fila afectada) and 'Advertencia: la máxima longitud de clave para un índice no agrupado es de 32 bytes. El índice no agrupado puede no funcionar correctamente si se supera este límite.' (1 fila afectada). The final execution time is 'Hora de finalización: 2026-09-21T09:25:06.7034364+02:00'.

Consulta de los datos json

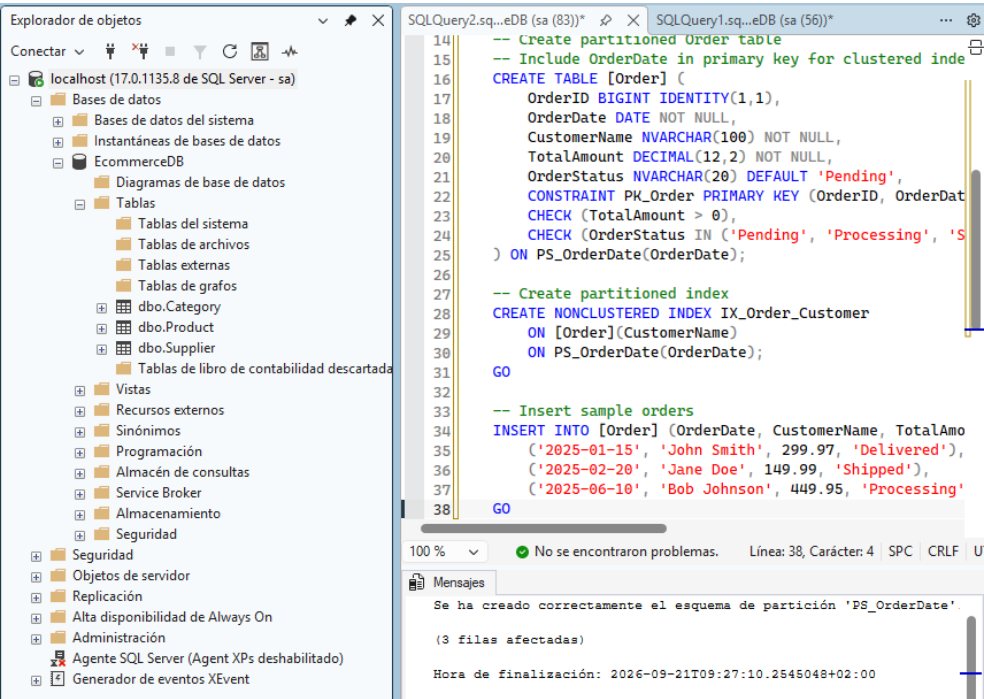
The screenshot shows the same SQL Server Enterprise Manager interface. The main window displays a query to retrieve JSON metadata data from the 'Product' table:

```
1 USE EcommerceDB;
2 GO
3
4 -- Query JSON data
5 SELECT
6 ProductID,
7 ProductName,
8 JSON_VALUE(Metadata, '$.color') AS Color,
9 JSON_VALUE(Metadata, '$.size') AS Size,
10 JSON_VALUE(Metadata, '$.material') AS Material
11 FROM Product
12 WHERE JSON_VALUE(Metadata, '$.color') = 'blue';
```

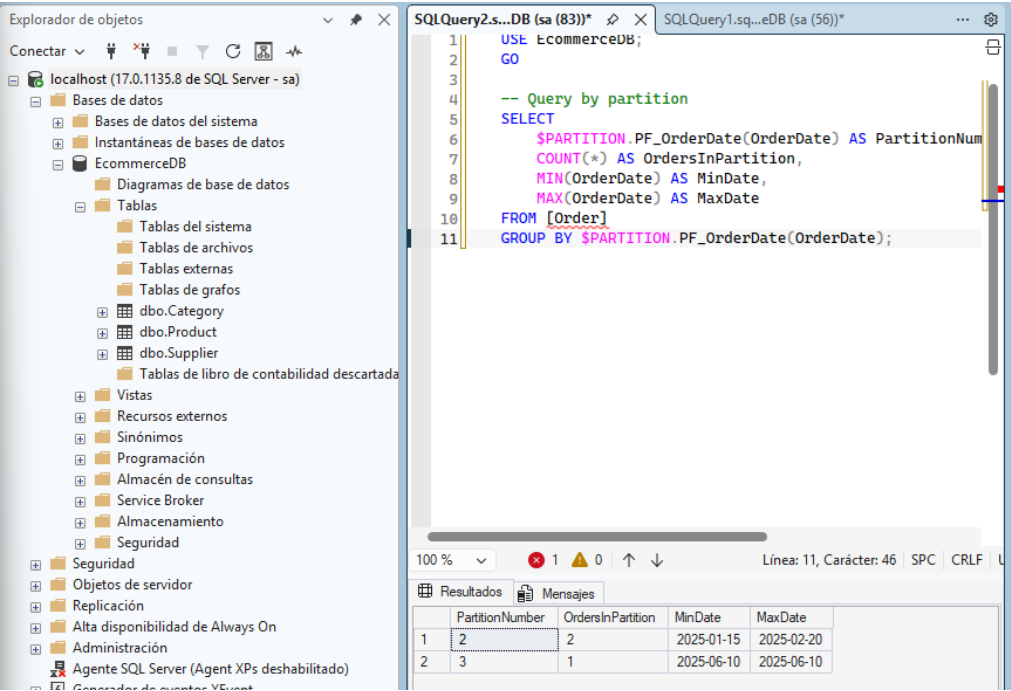
The status bar indicates 'Línea: 12, Carácter: 49'. The 'Resultados' (Results) pane at the bottom shows the query results in a table format:

ProductID	ProductName	Color	Size	Material
1	Wireless Mouse	blue	large	cotton

Crear una tabla de pedidos particionada



Consulta de la partición



Crear detalles del pedido con SEQUENCE

The screenshot shows the SQL Server Enterprise Manager interface on the left, displaying the database structure of 'EcommerceDB'. The right pane shows a SQL Query window with the following code:

```
7 INCREMENT BY 1;
8
9
10 -- Create OrderDetail table
11 CREATE TABLE OrderDetail (
12     OrderLineID INT PRIMARY KEY,
13     OrderID BIGINT NOT NULL,
14     OrderDate DATE NOT NULL,
15     ProductID INT NOT NULL,
16     Quantity INT NOT NULL,
17     UnitPrice DECIMAL(10,2) NOT NULL,
18     LineTotal AS (Quantity * UnitPrice),
19     CHECK (Quantity > 0),
20     CHECK (UnitPrice > 0),
21     FOREIGN KEY (OrderID, OrderDate) REFERENCES [Order]
22 );
23 GO
24
25 -- Insert order details using SEQUENCE
26 INSERT INTO OrderDetail (OrderLineID, OrderID, OrderDate,
27     ProductID, Quantity, UnitPrice)
28 VALUES
29     (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15',
30     1, 2, 99.99),
31     (NEXT VALUE FOR OrderLineSequence, 1, '2025-01-15',
32     2, 1, 149.99),
33     (NEXT VALUE FOR OrderLineSequence, 2, '2025-02-20',
34     1, 3, 99.99);
35 GO
```

The Messages pane at the bottom indicates that 3 rows were affected and provides the completion time: 2026-09-21T09:28:47.0492485+02:00.

Verificar los datos

The screenshot shows the SQL Query window with the following code:

```
1 USE EcommerceDB;
2 GO
3
4 SELECT * FROM OrderDetail;
```

The Results pane at the bottom displays the data inserted into the OrderDetail table:

OrderLineID	OrderID	OrderDate	ProductID	Quantity	UnitPrice	LineTotal
1	1	2025-01-15	1	2	99.99	199.98
2	1	2025-01-15	2	1	149.99	149.99
3	2	2025-02-20	1	3	99.99	299.97

Verificar objetos de la base de datos

Explorador de objetos

Conectar     

localhost (17.0.1135.8 de SQL Server - sa)

Bases de datos

Bases de datos del sistema

Instantáneas de bases de datos

EcommerceDB

Diagramas de base de datos

Tablas

Tablas del sistema

Tablas de archivos

Tablas externas

Tablas de grafos

dbo.Category

dbo.Product

dbo.Supplier

Tablas de libro de contabilidad descartada

Vistas

Recursos externos

Sinónimos

Programación

Almacén de consultas

Service Broker

Almacenamiento

Seguridad

Seguridad

Objetos de servidor

Replicación

Alta disponibilidad de Always On

Administración

Agente SQL Server (Agent XPs deshabilitado)

Generador de eventos XEvent

SQLQuery2.s...DB (sa (83))*

SQLQuery1.sq...eDB (sa (56))*

```
1 USE EcommerceDB;
2 GO
3
4 -- Verify constraints work
5 -- This should fail: negative price
6 INSERT INTO Product (ProductName, CategoryID, Supplier
7 VALUES ('Invalid', 1, 1, -50, 10);
```

100 % 6 0 Linea: 7, Carácter: 36 TABULACIONES

Mensajes

Mens. 547, Nivel 16, Estado 0, Línea 6
Instrucción INSERT en conflicto con la restricción CHECK 'CK_Proc
Se terminó la instrucción.

Hora de finalización: 2026-09-21T09:29:59.5014126+02:00

Verificar el json y las consultas

Explorador de objetos

Conectar     

localhost (17.0.1135.8 de SQL Server - sa)

Bases de datos

Bases de datos del sistema

Instantáneas de bases de datos

EcommerceDB

Diagramas de base de datos

Tablas

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Generador de eventos XEvent

SQLQuery2.s...DB (sa (83))*

SQLQuery1.sq...eDB (sa (56))*

```
1 USE EcommerceDB;
2 GO
3
4 -- Verify JSON queries work
5 SELECT ProductName, JSON_VALUE(Metadata, '$.color') AS
6 FROM Product
7 WHERE Metadata IS NOT NULL;
8
9 -- Verify partitioning
10 SELECT $PARTITION.PF_OrderDate(OrderDate) AS Partition
11 FROM [Order]
12 GROUP BY $PARTITION.PF_OrderDate(OrderDate);
13
14 -- Verify temporal table
15 SELECT ProductID, CurrentPrice, SysStartTime, SysEndTi
16 FROM ProductPrice FOR SYSTEM_TIME ALL
17 ORDER BY ProductID, SysStartTime;
```

100 % 12 0 Linea: 17, Carácter: 35 TABULACIONES

Resultados

Mensajes

ProductName	Color
1 Wireless Mouse	blue
2 Cotton T-Shirt	red

Partition	RecordCount
1	2
2	1

ProductID	CurrentPrice	SysStartTime	SysEndTime
1	109.99	2026-09-21 07:23:09.9330403	9999-12-31 23:59:59.9999999
2	149.99	2026-09-21 07:23:09.9330403	9999-12-31 23:59:59.9999999

